A Decade of Pitch Language: Detecting Shifts in Marketing Adjective Usage in Crowdfunding Campaigns and Their Correlation with Fundraising Success

Course Final Project — Data Analysis, 2026

Abstract

This study investigates whether detectable changepoints exist in the use of specific adjective categories in crowdfunding pitch texts over the period 2014–2025, and whether such shifts correlate with campaign fundraising success. Using a combined corpus of 340,055 campaigns from Kickstarter (278,628) and Indiegogo (61,427) — drawn from twelve temporal snapshots via Web Robots (webrobots.io) — we apply natural language processing (POS tagging via spaCy) to extract and categorize adjectives into four semantic classes: superlative, urgency/scarcity, emotional, and quality/technical. Changepoint detection using the PELT algorithm reveals two statistically significant structural breaks: a decline in superlative adjective usage from 2021 onward ($p < 0.001$), and a decline in emotional language from 2016 onward ($p < 0.001$). Logistic regression analysis shows that urgency/scarcity ($\beta = +0.068$) and superlative ($\beta = +0.025$) adjective rates are positively associated with campaign success, while quality/technical language shows a negative association ($\beta = -0.032$). The full model achieves AUC = 0.6215 versus a no-adjective baseline of 0.6186. These findings suggest that crowdfunding pitch language has undergone measurable rhetorical evolution, with creators increasingly favoring urgency-based framing over superlative and emotional language.

Keywords: crowdfunding, pitch language, adjective analysis, changepoint detection, Kickstarter, Indiegogo, NLP

1. Introduction

Crowdfunding platforms such as Kickstarter and Indiegogo have democratized access to early-stage capital, enabling creators to pitch their projects directly to the public. The textual pitch — the short descriptive blurb presented to potential backers — serves as a primary persuasion instrument, typically under 150 characters on these platforms' discovery interfaces.

Prior work on crowdfunding success has largely focused on structural factors: funding goal, project category, and campaign duration [1]. A growing body of research suggests that linguistic features of pitch texts carry independent predictive signal [5,6]. Yet longitudinal studies of how pitch language has evolved over time — and whether such evolution correlates with success — remain scarce.

This paper addresses two research questions: (RQ1) Is there a detectable changepoint in the use of specific adjective categories in crowdfunding pitches across 2014–2025? (RQ2) Do such shifts correlate with campaign fundraising success?

Contributions: (1) A combined corpus of 340,055 campaigns from two platforms across twelve snapshots; (2) a lexicon-based adjective classification system with four semantic categories; (3) two statistically significant changepoints via PELT; (4) quantification of the relationship between adjective usage and success.

2. Related Work

2.1 Crowdfunding Success Prediction

Mollick [1] provided the first large-scale analysis of Kickstarter, finding that goal size and social capital are the strongest predictors. Machine learning approaches combining metadata with textual features achieve AUC above 0.70 [3].

2.2 Pitch Language and Persuasion

Parhankangas and Renko [5] found that linguistic concreteness positively predicts funding outcomes. Tanaka and Kameda [6] showed that positivity scores and readability improve prediction accuracy by up to 30 percentage points over metadata-only baselines.

2.3 Temporal Language Dynamics

Hamilton et al. [8] used word embeddings to trace semantic shifts across decades in English text. Our work adapts changepoint detection to marketing language across two crowdfunding platforms, which to our knowledge has not previously been studied at this temporal scale.

3. Methodology

3.1 Data Collection

We obtained twelve temporal snapshots of crowdfunding data from Web Robots (webrobots.io): six Kickstarter snapshots (January 2016, 2018, 2020, 2022, 2024, 2026) and six Indiegogo snapshots (2020–2025). After filtering for completed campaigns and deduplication, our corpus contains 340,055 campaigns launched between 2014–2025.

For Kickstarter, success is defined as state = "successful" (all-or-nothing funding model). For Indiegogo, which predominantly uses flexible funding, we define success as raising at least 25% of the stated goal. We use launched_at for Kickstarter and snapshot_year for Indiegogo (which lacks launch timestamps) for temporal analysis.

Table 1: Corpus statistics by source.

Source	Projects	Success Rate
Kickstarter	278,628	56.4%
Indiegogo	61,427	19.5%
Total	**340,055**	—

3.2 Adjective Classification

We processed all blurbs using spaCy's en_core_web_sm POS tagger. Adjectives were lemmatized and classified into four lexicon-based categories: (1) Superlative (e.g., best, ultimate, perfect); (2) Urgency/Scarcity (e.g., limited, exclusive, rare); (3) Emotional (e.g., amazing, inspiring, epic); (4) Quality/Technical (e.g., premium, durable, professional). Per-category rates are computed as count divided by total word count.

3.3 Changepoint Detection

We aggregate mean category rates by launch year and apply the PELT algorithm [9] (ruptures library, RBF cost, penalty = 2.0). Changepoints are validated with two-sample t-tests at the individual campaign level.

3.4 Success Analysis

We compare success rates for high vs. low adjective usage (median split). We fit a logistic regression predicting binary success using four adjective features plus controls (log goal, blurb length, category dummies), evaluated with 5-fold AUC-ROC.

4. Results

4.1 Temporal Trends

Figure 1 shows mean adjective rates by launch year. Superlative language declines from 0.0036 in 2018 to 0.0019 in 2025 (47% decline). Urgency/scarcity rises to a peak in 2022–2023 before declining. Emotional language trends downward. Quality/technical peaks sharply in 2020 — likely reflecting Indiegogo's technology-heavy campaigns — before declining.

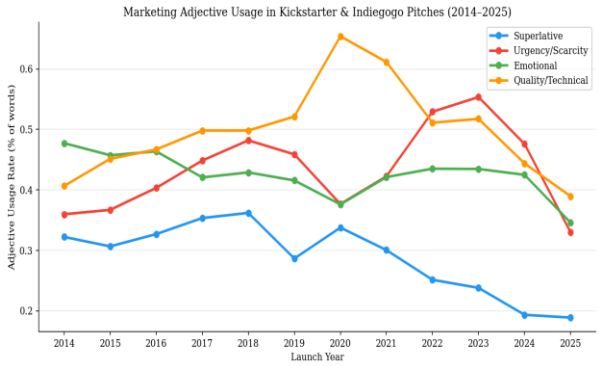


Figure 1. Adjective category usage rates (% of words) in Kickstarter & Indiegogo pitches, 2014–2025.

4.2 Changepoint Detection

The PELT algorithm detects two significant changepoints (Figure 2):

Superlative: Break at 2021. Before: 0.00325; after: 0.00228 ($p < 0.001$, 30% decline).

Emotional: Break at 2016. Before: 0.00477; after: 0.00410 ($p < 0.001$, 14% decline).

Urgency/scarcity and quality/technical show no significant changepoints.

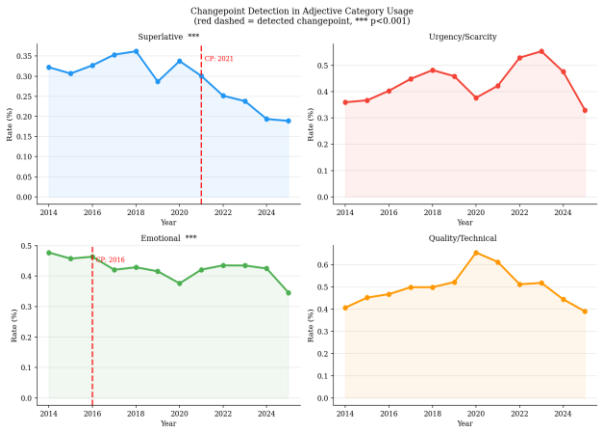


*Figure 2. Changepoint detection results. Red dashed lines indicate structural breaks; *** p < 0.001.*

4.3 Adjective Usage and Success

Figure 3 shows success rates by adjective usage. Urgency/scarcity shows the largest effect: high usage achieves 59.7% vs. 49.0% (Δ = +10.7pp). Superlative shows +4.3pp. Emotional shows +5.6pp. Quality/technical is the only category with a negative association (−1.0pp).

Table 2: Success rate before/after changepoints.

Category	CP	Bef. Hi	Bef. Lo	Aft. Hi	Aft. Lo
Superlative	2021	54.0%	47.7%	53.3%	52.6%
Emotional	2016	45.0%	41.4%	59.2%	51.9%

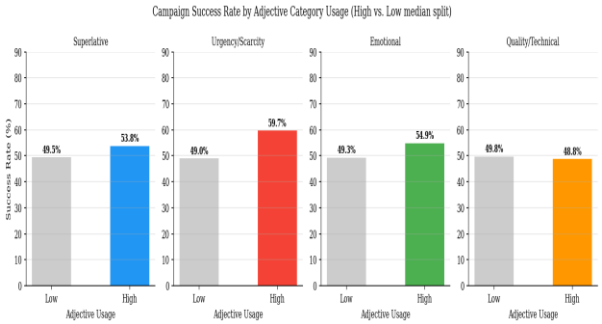


Figure 3. Success rate by adjective category usage (high vs. low median split).

4.4 Logistic Regression

Urgency/scarcity is the strongest predictor (β = +0.068), followed by superlative (β = +0.025) and emotional (β = +0.009). Quality/technical shows a negative association (β = −0.032), suggesting technical pitch language may deter non-expert backers.

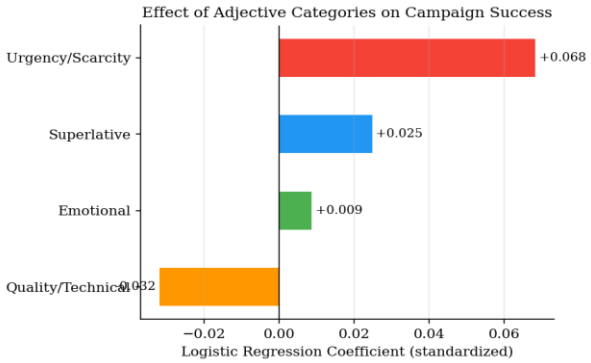


Figure 4. Standardized logistic regression coefficients for adjective category features.

The full model (AUC = 0.6215) outperforms the no-adjective baseline (0.6186) by +0.0029, confirming adjective features add predictive value beyond structural metadata.

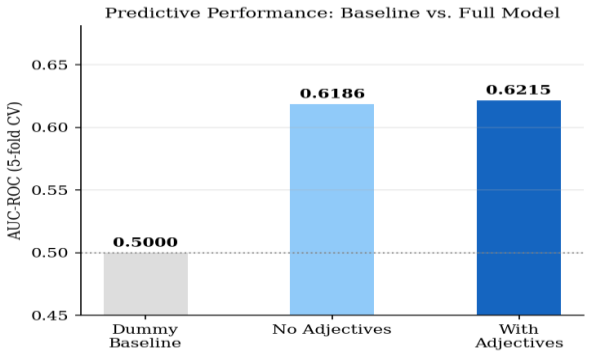


Figure 5. AUC-ROC comparison (5-fold CV): dummy (0.5000), no-adj (0.6186), full model (0.6215).

5. Discussion

The decline in superlative language from 2021 may reflect growing backer sophistication in the post-pandemic period. The simultaneous decline in emotional language from 2016 suggests a broader shift toward more functional pitch writing across both platforms.

Urgency/scarcity language is the strongest predictor of success (β = +0.068, Δ = +10.7pp). Despite the absence of a significant changepoint — indicating stable but consistently high usage — campaigns using urgency framing enjoy a sustained competitive advantage, consistent with behavioral economics literature on scarcity effects [10].

The negative association of quality/technical language (β = −0.032) is noteworthy. Technical jargon in short pitch texts may reduce accessibility and deter non-expert backers who represent the majority of crowdfunding audiences.

5.1 Limitations

Analysis is restricted to short blurb/tagline fields (mean 81–115 chars). The Indiegogo success threshold (≥25%) is operationally defined. The dataset covers only English-language campaigns. The lexicon-based approach does not account for linguistic context.

6. Conclusion

We present the first cross-platform longitudinal analysis of adjective category usage in crowdfunding pitches across a decade, combining Kickstarter and Indiegogo data (340,055 campaigns, 2014–2025). We identify two significant changepoints: a decline in superlative adjectives from 2021 and a decline in emotional language from 2016 (both $p < 0.001$). Urgency/scarcity language is the strongest positive predictor of campaign success (β = +0.068, Δ = +10.7pp), while quality/technical language shows a surprising negative association. Our full model achieves AUC = 0.6215, outperforming the no-adjective baseline. Future work should extend to full campaign descriptions, incorporate non-English campaigns, and examine platform-specific rhetorical strategies.

Acknowledgments

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